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Raspberry Pi Laparoscope

Mechatronic Project 478
Progress Report

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2026

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Declaration

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Institutional Acknowledgement

This declaration aligns with Stellenbosch University's Position Statement on the ethical use of AI in Research and Teaching-Learning-Assessment¹, the Draft interim SU guidelines on allowable AI use and academic integrity in assessment², and draws on international best practices.

¹[Position Statement on Ethical use of Artificial Intelligence in Research and Teaching-Learning Assessment](#)

²[Draft interim SU guidelines on allowable AI use and academic integrity in assessment](#)

Executive summary

Title of Project
Raspberry Pi Laparoscope
Objectives
The objective of this project is to design, program, build and test a low-cost laparoscope.
What is current practice and what are its limitations?
Current laparoscope use in industry is expensive and not feasible to purchase for training and prototype testing of new laparoscopic medical robots.
What is new in this project?
For this project, instead of specialized lenses and camera modules, off-shelf products will be utilized to develop the product.
If the project is successful, how will it make a difference?
If this project is successful, it can save cost when designing and prototyping new medical robots. It can be utilized for training of new surgeons trainees.
What are the risks to the project being a success? Why is it expected to be successful?
The main risk of this project is delayed due to unavailability of hardware components. Therefore, the ordering of products is the main priority during early stages of the project.
What contributions have/will other students made/make?
Another student is performing the same project as me, thus notes and methods will be compared.
Which aspects of the project will carry on after completion and why?
The developed laparoscope will be utilized in future projects, where medical robots will be developed. The test station will also be utilized in testing future laparoscope designs.
What arrangements have been/will be made to expedite continuation?
To prevent delays, safety measures have been taken to prioritise the ordering of important hardware components.

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Chapter 1

Introduction

1.1 Background

The Hippocratic Oath is a fundamental principle that guides decisions and advancements in the medical profession. It motivates doctors and physicians to make ethical decisions to minimize patient harm. It also drives physicians to continuously improve practices to reduce surgical risks and complications. This principle has driven the discovery of minimally invasive surgery (MIS). (Ibrahim *et al.*, 2021)

Minimally invasive surgery has become the standard practice across surgical fields and is a rapidly growing division in the medical industry. One branch of this field is laparoscopy, which is used to perform advanced and complex surgical procedures. (Ibrahim *et al.*, 2021)

Laparoscopy is surgical approach where small incisions are made to allow insertion of specialized equipment into the patient's body. The primary instrument is the laparoscope, which is typically paired with other thin rod equipment based on the operations demands. Compared to traditional open surgery, laparoscopy offers advantages including faster recovery rates and reduced morbidity and patient discomfort, though it requires a steeper learning curve. These benefits translate to cost savings through shorter hospital stays and reduced analgesic requirements. (Subido *et al.*, 2018)

The laparoscope is a fundamental instrument to laparoscopic surgery. It consists of a thin rod with an attached camera that is inserted into the patient's body. The captured footage is displayed in real-time to the surgical staff on a monitor. Based on the visible footage, the surgical staff make diagnosis, perform procedures, and coordinate instrument positioning based on the visual feedback. (Ibrahim *et al.*, 2021)

Recent advancements in laparoscopy focus on improving the laparoscope's visual capabilities through several approaches. These approaches include increasing the camera's field of view, implementing three-dimensional (3D) visualization, and incorporating artificial intelligence (AI) or machine vision technologies to captured footage. Additional advancement made to modern laparo-

scopes is the utilization of robotic manipulators, with 3 or more degrees of freedom (DOF) to assist the doctor during operations. (Subido *et al.*, 2018)

One of the main challenges of laparoscopic surgery is that it is a technically complex surgery to perform. Thus, more training is required by inexperienced surgeons. However, laparoscopes are expensive equipment which makes them inaccessible for individual practitioners or small hospitals to purchase. (Xiao *et al.*, 2014)

Additionally, Stellenbosch University host a open or career day, where potential students can attend to see which possible degrees they can study at the university. Ideally, the university would appreciate to have a working laparoscope, to demonstrate how laparoscopic surgeries are performed. Consequently, the laparoscope will be used by the attending students to experience how laparoscopic surgeries are performed on phantom bodies.

Therefore, there is a need for a low-cost laparoscope, which will be used for training and demonstration purposes and not on living patients.

1.2 Objectives

The aim of this project is to design, build and test a low-cost laparoscope.

This project's main objectives are to:

- Design, build and program a functional laparoscope by incorporating various Raspberry Pi components.
- Design a performance test to assess the laparoscope functionality and image quality abilities.
- Evaluate the developed laparoscope performance after performing the test.

1.3 Scope

To developed the laparoscope, there are various aspects that needs to be considered. These aspects range from physical properties to software properties and all have various scopes. However the scope for this project is subject to change based on the timeline of the project or change of the desired objectives of the project.

An important scope boundary is that the developed laparoscope, should not meet all medical grade standards. Since, it will not be used on living patients, the product should not be made out of medical grade equipment or materials. However, it needs to function similarly to medical grade laparoscopes, henceforth, should meet certain functionality standards for laparoscopic equipment.

Additionally, the developed laparoscope does not require to have any degrees of freedom (DOF). The viewing angle, angle between the optical axis of

the camera and the laparoscope axial axis, should aimed to be 30 degrees. It is not required to incorporate a DOF to the viewing angle of the laparoscope.

To determine the laparoscopes capabilities and properties a testing station is required. Since Stellenbosch University does not have a testing station for laparoscopic products, it is required to design and build one, as well as articulate the testing procedures for the laparoscope.

Considering the hardware components of the laparoscope, it is not within the project scope to manufacture hardware components from first principles. Rather, available hardware components will be purchased and repurposed if necessary to meet the project requirements.

Considering software of the laparoscope, it is required that system should be able to only operate the different modules. Any additional software application, such as machine learning or machine vision, is not within the projects scope.

1.4 Motivation

As mentioned in the section 1.1, the main challenge with laparoscopic surgery is that it is a technically more complex procedure to perform, compared to open surgery. Thus, inexperienced surgeons require more training with the laparoscope and its additional equipment to achieve proficiency. However, laparoscopes are expensive surgical equipment, that isn't affordable investment for individuals or small medical facilities.

This project could also add value to the development of new medical robots. Most preliminary designs of medical robots require laparoscopes that can test the robot's capabilities and flaws. However, they do not require expensive medically approved laparoscopes, rather, low-cost alternatives are preferable, as they minimize financial loss if damaged during testing or development.

Additionally, a low-cost laparoscopes can be used during future career days to demonstrate how laparoscopic surgeries can be performed for prospective students.

1.5 Use of AI Tools

Anthropic is the main Artificial Intelligence (AI) company that will be used for this project, specifically its Claude Sonnet 4.5 model. This AI model assisted with research papers identification, sentence construction, grammar checks and code debugging. All AI-generated material will be reviewed and verified by the student to ensure the integrity and originality of the provided work.

1.6 Proposal Overview

This proposal aims to enlighten the reader of the various sections of work required to perform this project. Chapter 2 informs of the history of laparoscopy, afterwards it explains various segments of the work required to understand to complete the project. Chapter 3 aims to provide an overview of the entirety of the project work procedures. It describes the research methodology utilized; the workplan; the resources required; the expected results; the schedule; and the risks associated with this project. The last chapter of this proposal, chapter 4, provides a final motivation why this project should be performed.

Chapter 2

Literature Review

A literature review has been conducted for this project. This section of the report documents the relevant research performed and divides into the following sections:

- **Laparoscopy Overview:** An overview of what laparoscopy is and its history.
- **Current Technological Advancements:** The current directions that modern laparoscopy is improving on.
- **Current Available Technology:** A summary of current Laparoscopes that is available in South Africa.
- **Available Hardware Components:** An overview of different hardware modules available for this project.
- **Software Considerations:** An brief overview of the available programs that can be used for this project.
- **Visual Inspection Considerations:** An overview of visual inspection and aspects of camera that need to be mastered.

2.1 Laparoscopy Overview

Laparoscopy is a branch of the endoscopic medical field. It has become an integral part of surgical procedures in all field in the recent decades. Inspired by the minimally invasive surgery (MIS) approach, laparoscopic surgery consist of making 1 or more small incisions on the abdomen of the patient. Surgeons use these incision to insert thin rod medical equipment into the patient to perform the surgery. (Ibrahim *et al.*, 2021)

An essential tool of this surgical procedure is the laparoscope. Due to small size of the incisions, the surgeons' view is limited. The laparoscope is the solution to this problem. It allows the surgeons to view what is occurring within

the patient while the operation is being performed. Therefore, they can clearly manipulate their equipment from outside the patient to perform complex procedures inside the patient. (Subido *et al.*, 2018)

Modern laparoscopes are equipped with small cameras and light sources at the end of the tool. This camera is inserted into the patient and the captured footage is displayed in real-time to the surgical staff with the means of a monitor. With the visible footage, the surgical staff can make diagnosis and perform operations. (Subido *et al.*, 2018)

However, the first laparoscope has existed for almost the last century. They didn't consist of cameras, rather, out of complex lens assemblies at the end of the laparoscope and cold light sources. They were used in a similar method as a microscope. However, this severely limited the surgeon's ability to operate as well as limited the surgical staff ability to assist the head surgeon. (Ibrahim *et al.*, 2021) Figure 2.1 illustrates how the original laparoscope was designed.

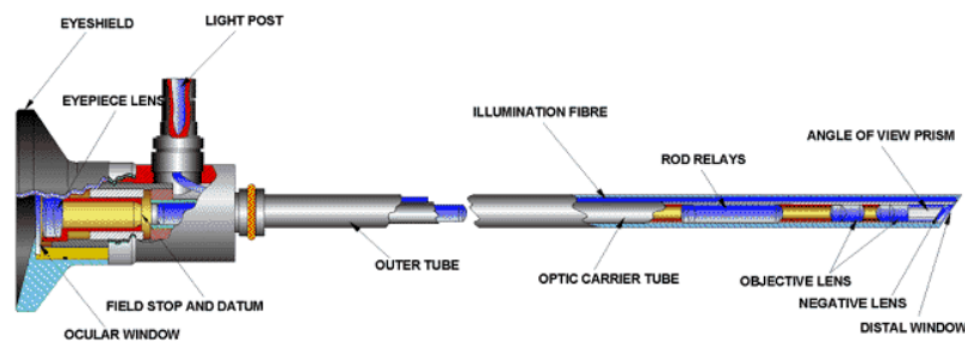


Figure 2.1: Original laparoscope design. (ebme.co.uk, 2026)

Laparoscopic surgery has been one of the greatest advancements in medical industry in recent years. Buia *et al.* (2015) shows that laparoscopic surgery has quicker recovery time and less pain after surgery compared to traditional open surgery. However, it shows that certain procedures are more time-consuming and more complex to perform.

2.2 Current Technological Advancements

Since the addition of cameras to laparoscopes, there has been a large addition of advancement available. These advancement range from hardware improvements to additional software programming abilities to the existing design.

The main hardware advancement made to laparoscopes is the improvement of the camera modules. However, additional advancements has been made with the addition of robotics to the laparoscopes, which has greatly assisted the laparoscopic surgical industry. Robotic manipulators are utilized to assist surgeons by maintaining stable control of the laparoscope, therefore, ensuring consistent and clear visualization during the procedure. (Subido *et al.*, 2018)

There are multiple additions currently being made to the software of laparoscopes. *Subido et al. (2018)* list the following additions:

- Digital imaging enhancement techniques.
- AI incorporation.
- 3D visualization.
- Object tracking.

2.3 Current Available Technology

There is a range of different laparoscope or endoscope suppliers. Each of these suppliers produce more than 1 product. Therefore, only suppliers for South African hospitals will be reviewed and Intuitive Surgical that produces the *Da Vinci* system, that has been dominating the surgical robot market. (*Subido et al., 2018*)

One of the main suppliers of laparoscopic equipment in South Africa is Karl Storz Endoskope, a German company, with a sales office in Cape Town. They produce various products in the medical industry. (*karlstorz.com, 2026a*) One of their products is the *TIPCAM1 RUBINA*, which can be seen in figure 2.2 below.

TIPCAM1 RUBINA is a rigid endoscope with the ability of 0° and 30° viewing angle. It also comes with automatic horizon control and the ability to easily swap between 2D and 3D imaging. Its has an outer diameter of 10.3 mm and a working length of 320 mm. It is has 4K image resolution and has a mass of 420 g. (*karlstorz.com, 2026b*)



Figure 2.2: Figure of the TIPCAM1 RUBINA. (*karlstorz.com, 2026b*)

Olympus Corporation is another company that supplies medical equipment globally. They are a Japanese manufacturing company that produces the *ENDO-EYE 3D 10 mm*, which can be seen in figure 2.3 below. The *ENDO-EYE 3D 10 mm* is a video telescope, with the ability of 3D imaging at a viewing angle of 0° and 30°. It is able to perform image rotation at 30° viewing angle, while maintaining the horizon. (*olympus.com.au, 2026*)



Figure 2.3: Figure of the ENDOEYE 3D 10 mm. (olympus.com.au, 2026)

Certain international companies use local distributors to sell their products. Intuitive Surgery, an American company, uses this method. They designed and manufacture the *Da Vinci* medical robot. This system has various variants for different surgical purposes. Currently, there is *Da Vinci Xi* and *Da Vinci X* for multi port surgery. However, the *Da Vinci SP* is designed for single port surgery. These robot systems are equipped with 7 DOF movement and 3D visualization. Their only drawback is their large size and enormous financial procurement cost. (intuitive.com, 2026) Below in figure 2.4 of the *Da Vinci SP*.



Figure 2.4: Figure of the Da Vinci SP. (intuitive.com, 2026)

2.4 Available Hardware Components

There are multiple hardware components required to build a laparoscope. Therefore, this section will be divided into multiple subsections where different prod-

ucts will be reviewed for each component. These subsections include:

- Single board computer module.
- Camera module.
- Lighting module.

2.4.1 Single Board Computer Module

Single board computers (SBC) is a credit-card-size computer, able to perform complex computations while operating various systems. For the laparoscope, it will be the main "brain" of the system. Therefore, it is critical that it can perform all necessary tasks it requires. There is 2 main suppliers of SBC:

- Raspberry Pi Ltd.
- Arduino AG.

Raspberry Pi has a large range of products that is available for the project. However, Raspberry Pi's SBCs has an important feature installed into most of SBC, which would be beneficial to the project. This feature is the designated camera serial interface (CSI) or the MIPI camera display transceivers, according to Raspberry Pi. This feature allows for high-quality videos at high speeds. (raspberrypi.com, 2026a) *Table 2.1* below show the available options of all the Raspberry Pi SBC:

Table 2.1: Raspberry Pi SBCs.(raspberrypi.com, 2026b)

Raspberry Pi SBC	RAM	CPU Speed	MIPI Ports	Cost
Pi 5	1-16 GB	2.4 GHz	2	R 3 266.00
Pi 4 Model B	1-8 GB	1.5 GHz	1	R 1 360.00
Pi 3 Model B+	1 GB	1.4 GHz	1	R 640.43
Pi Zero 2 W	512 MB	1 GHz	2	R 240.16
Pi Zero 2	512 MB	1 GHz	2	R 309.90

All of the SBC listed in *Table 2.1* above have these features:

- 40 GPIO Pins.
- Wi-Fi Capabilities.
- Bluetooth Capabilities.
- USB Ports.

Arduino, similar to Raspberry Pi, have a wide range of products available for this project. However, they have one absent feature, namely the lack of CSI feature. Therefore, they are incompatible with the Raspberry Pi Camera Modules. However, they can use other camera modules, with the use of SPI and I2C GPIO pins. Although this options isn't favourable, due to trade-off in quality during longer videos, it is still a feasible option that can be explored. (Public, 2022) The *Table 2.2* below shows the available products from Arduino:

Table 2.2: Arduino SBCs. (store.arduino.cc, 2026)

Arduino SBC	RAM	CPU Speed	SPI Ports	Cost
UNO R4 WiFi	32 kB	48 MHz	1	R 488.46
UNO R4 Minima	32 kB	48 MHz	1	R 352.33
UNO R3	2 kB	16 MHz	1	R 469.24
GIGA R1 WiFi	1 MB	480 MHz	2	R 1 247.58
Nano ESP32	512 kB	240 MHz	1	R 358.74
Nano RP2040 Connect	264 kB	133 MHz	4	R 389.17
UNO Q	2 GB	2 GHz	1	R 1 035.54
Nano 33 BLE	256 kB	64 MHz	4	R 456.43
Nano 33 IoT	520 kB	240 MHz	6	R 427.6
Nano Every	6 kB	20 MHz	1	R 232.22

2.4.2 Camera Module

The camera module is an important component for this project that requires careful considerations when selecting. It is the main component that will be inserted into a patients body to capture the real-time images for the surgical staff to perform operations. Therefore, the quality of the camera module will dictate the performance of the entire laparoscope.

The camera module has the most specifications regarding physical properties out of the components. The entire module must fit within a $\varnothing 10$ mm hole. This proved to be the largest failing criteria for camera module available on the market, due to large camera assemblies; or the large printed circuit board (PCB) attached to the camera assemblies. *Figure 2.5* below demonstrates how the PCB attached to the camera module is to large fit inside the $\varnothing 10$ mm hole. (arducam.com, 2020)

The second most common failing criteria for camera modules were the focus depth of the camera modules. Since the distance from the camera module lens and focus object is small, 5 mm to 100 mm, camera modules with large focus depth ranges are not practical for this project. Therefore, eliminating products from the hummingbird mobile camera module series, developed by Leopard Imaging, (leopardimaging.com, 2026) and the OV5647 spy camera module, developed by ArduCam. (store.arduino.cc, 2026)



Figure 2.5: Figure of Raspberry Pi Camera Module 3. (raspberrypi.com, 2026a)

Based on these 2 main selection criteria, 2 products have been selected to use for this project:

- DF Robot 3-in-1 endoscope camera, produced by DF Robot.
- 1 MP 3.5 mm OV9734 HD endoscope medical camera module, produced by Vision Meta.

Table 2.3 compares the 2 product mentioned above, while *figure 2.6a* and *figure 2.6b* illustrate the products.

Table 2.3: DF Robot compared to Vision Meta.

Feature	DF Robot	Vision Meta
Diameter (mm)	8	3.5
Field of View (°)	70	90
Depth Focus (mm)	N/A	5 - 100
Resolution	0.864 MP	0.922 MP
Built-in LED	Yes	No
Cost	R 457.70	R 995.96 (\$ 60.00)
Shipping Cost	R0.00	R 1 344.55 (\$ 81.00)



(a) DF Robot endoscope module. (robotics.org.za, 2026)



(b) Vision Meta endoscope module. (visionmeta.com, 2026)

Figure 2.6: Illustration of camera modules

These camera module have USB compatibility and meets the physical specifications for this project and would in theory work. However, a feature that laparoscopes have is that the viewing angle is 30° . Due to the DF Robot endoscope's large diameter and length, this feature makes it unpractical for this project, unless the module is modified.

2.4.3 Lighting Module

Due to the laparoscopes being inserted into patient's bodies, there will be limited illumination within the patient, which will impact image quality. This can be easily resolved by installing illumination hardware to the laparoscope. These hardware components should be small in design, while producing enough illumination, which is measured in luminous intensity (cmd). Micro Robotics, has 2 main products that can be used for this project. Both products are listed in Table 2.4 and illustrated in figure 2.7 below. (robotics.org.za, 2026)

Table 2.4: List of illumination hardware. (robotics.org.za, 2026)

Product	Luminous Intensity	Dimensions
LED SMD 1206	200 - 850 cmd	3.2 x 1.6 x 1.1 mm
LED SMD 0603	45 - 180 cmd	1.6 x 0.8 x 0.55 mm

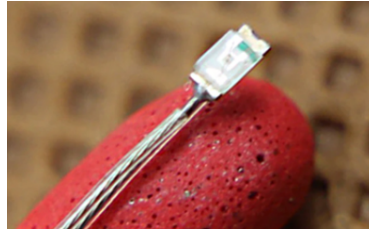


Figure 2.7: Figure of LED SMD 0603.(robotics.org.za, 2026)

2.5 Software Considerations

The most important software application of this project is that of the camera module interface. Since, both camera modules that are going to be used in this project, uses USB interface, the recommended software to operate the camera modules is AMCap. (robotics.org.za, 2026)

The software language is an important aspect that needs to be considered. However, this is dependent of the SBC used. If a Raspberry Pi is used, then Python is recommended language to use. (raspberrypi.com, 2026a) However, if an Arduino is used then the recommended language to use is either C or C++. (docs.arduino.cc, 2026) Visual Studio Code (VS Code) can be used to perform all programming for this project.

2.6 Visual Inspection Considerations

This project requires a camera to be a functional laparoscope. However, there is various factors to be considered to take high-quality images. This section of the literature reviews aims to explain each of these factors. Consequently, how they can impact this project.

Important concept to understand, is how a digital image is captured. A camera sensors consists of a matrix of millions of light cavities. When a picture is being taken, these cavities, called "photosites", are exposed to the photons of the object or scene. The photons are then collected by the photosites and stored as an electrical charge. Consequently, when the image is finished being taken, these charges are quantified and then processed to make the image. (cambridgeincolour.com, 2020b)

One aspect that needs to be mastered to captured quality images is the exposure triangle. It is 3 pillars that dictates the quality of an image. The first pillar, called aperture, determines the size of the area opened in the lens which light can filter through to sensor. The larger the aperture value, measured in f-stop values, the smaller the hole size where light filters through the lens. This pillars relates to the depth of field of the images. The smaller aperture correlates to shallower depth of field images. (cambridgeincolour.com, 2020a)

Shutter speed is the second pillar of the exposure triangle. When a digital image is being taken the lens opens up to expose the photosites to the light. Consequently, shutter speed refers to total duration of the lens being open. If the shutter speed increases, then the sensor is exposed to reduced duration of light from the images. Therefore, as the shutter speed increases, the images will gain more quality, with more definitive objects that are more "frozen" in the image. (cambridgeincolour.com, 2020a)

The third pillar is ISO. This determines how sensitive the sensor is to the exposed light. As the ISO value increases, the higher the sensitivity value will become. Therefore, the image will have increased levels of noise captured if the ISO value is increased. (cambridgeincolour.com, 2020a)

Focal length is a critical feature that has to be considered for this project. Focal length, is the distance between the camera sensor and the camera lens. As this value increases, then the field of view value will decrease. Field of view, measured in degrees, is essentially how much of an image can be seen. If this value increases, then a wider image can be taken. (Morrow, 2026)

Chapter 3

Content Chapter

3.1 Research Methodology

For this project, a design research methodology will be utilized to solve the problem. Since this project requires a design research methodology to be completed, there is a structure that can be followed. The initial phase that needs to be completed is a design requirement phase. In this section it required to determine:

- Stakeholders;
- Design independent parameters (DIP);
- Boundary between wider system of interest and narrower system of interest;
- Functional Requirement;
- Non-functional Requirements;
- Technical performance measures (TPM);

of the system. It is essential to remain neutral in this section.

The second phase that needs to be completed is the concept development phase. In this section, concepts are developed and described to meet all the requirement in the design requirement phase. These concepts are evaluated before the best concept or combination of concepts is selected. The selected concept(s) is refined further, before a final evaluation is performed.

The last 2 sections of this methodology is the detail design and analysis phase; which is then followed with the evaluation phase. The detail design and analysis phase requires the researcher to utilize the refined concept from the preceding section and create a detail design of the concept. This process does require an iterative process to create the final design. A drawing pack is required at the end of this phase that will be evaluated. The evaluation phase

consist of evaluating the developed product. This includes building the product, before it can be tested. Based on the test results of the product, an evaluation can be performed.

3.2 Workplan

This section of the proposal covers the planned workplan that will be conducted for this project. As established in the prior *section 3.1*, this project will follow a design research methodology. Therefore, it can be assumed that the project will require these following activities:

3.2.1 Literature Review

Study the literature of laparoscopic surgery to gain an understanding of the conditions that a laparoscope will experience. Consequently, there will be required to study literature that will inform what attributes the developed laparoscope should satisfy, to be used for testing purposes on future projects.

3.2.2 Component Review

A detailed investigation was conducted to create a list of possible hardware components, which can be used to integrate within the developed product. The hardware components that were investigated are:

- Single board computers.
- Camera modules.
- Illumination modules.
- Display modules.

Except for the camera modules, only local supplier stock were investigated for this project, since they have adequate products available. Except for the DF Robot endoscope, mentioned in *subsection 2.4.2*, no other products were found to meet the physical demands in south african suppliers for the camera modules. Henceforth, the reason why international companies were investigated for possible solutions for the camera module of the project.

3.2.3 Compile Design Requirements for Laparoscope

As mentioned in *section 3.1*, there are a list of requirements that needs to be determined. These requirements are all listed in the tables below, as well as shown in the diagrams below. The stakeholder analysis is shown in *table 3.1* below:

Table 3.1: Stakeholder Analysis

Stakeholder ID	Stakeholder	Classification	Description	Life Cycle Stages	Key Interest/Needs
SH_01	Dr Diayar	Indirect	The doctor that recommend the development of the system	Concept	The functional development of the system
SH_02	Prof Schreve	Secondary	Supervisor of the project	All stages	The functional development of the system, and the engineering design methodology followed to develop the system
SH_03	Medical trainee	Primary	The personal that will use the developed system to improve their surgical abilities	Operation	The low-cost production of the system and the easy operations of the system
SH_04	Hardware suppliers	Secondary	The suppliers of the hardware components for the system	Design, Installation	Technical specifications of the hardware components
SH_05	Structural manufacturers	Secondary	Personnel responsible for the construction of structural components of the system	Design, Installation	Technical specifications of the structural components

Continued on next page...

Stakeholder ID	Stakeholder	Classification	Description	Life Cycle Stages	Key Interest/Needs
SH_06	Stellenbosch University	Secondary	Main financial supplier and main management team of the project	All stages	The cost of the project, academic procedures followed and safe operations during all stages of the project
SH_07	Future researchers	Primary	Future personal that will use the developed system or select components of the system to perform further researchers	Maintenance, Operation	The technical conditions of the developed system, the technical conditions of systems hardware components
SH_08	Construction and maintenance staff	Secondary	Personnel responsible for the construction and maintenance for the entire system	Installation, Maintenance	System design and building procedures, systems maintenance procedures
SH_09	Design team	Secondary	Personnel responsible for the design of the entire system	Design	System requirement and hardware physical properties
SH_10	Software team	Secondary	Personnel for the software component of the entire system	Design, Installation	System requirements and hardware software capabilities

Continued on next page...

Stakeholder ID	Stakeholder	Classification	Description	Life Cycle Stages	Key Interest/Needs
SH_11	Potential student attendees	Primary	Potential students that will use the system to perform surgical procedures	Operation	Easy operations and usability of the system

The DIP requirements for the laparoscope can be seen in *table 3.2* below. Additionally, how the DIPs interface from the narrower system of interest (NSOI) with the wider system of interest (WSOI) and environment can be seen in *figure 3.1* below:

Table 3.2: Design Independent Parameters

DIP_ID	Design Independent Parameter Description	Value / Range	Reference
DIP_01	Laparoscope maximum diameter	$\varnothing < 10$ mm	Supervisor
DIP_02	Grid electricity availability	230 V, 15 Ampere at 50 Hz	worldstandards.eu
DIP_03	Ambient operating temperatures range (Stellenbosch)	2 C° - 42 C°	weather-and-climate.com
DIP_04	Laparoscopes total building cost	R 6000.00	Stemlearn (2026)
DIP_05	Laparoscope illuminance produced	40 000 LUX - 160 000 LUX	IEC 60601-2-41:2021
DIP_06	Relative maximum humidity in operating rooms	< 81%	weather-and-climate.com
DIP_07	Maximum deviation of specified viewing angle	10°	ISO 8600-1:2025
DIP_08	Frequency of real-time digital images updates	30 Hz - 60 Hz	(karlstorz.com, 2026b)

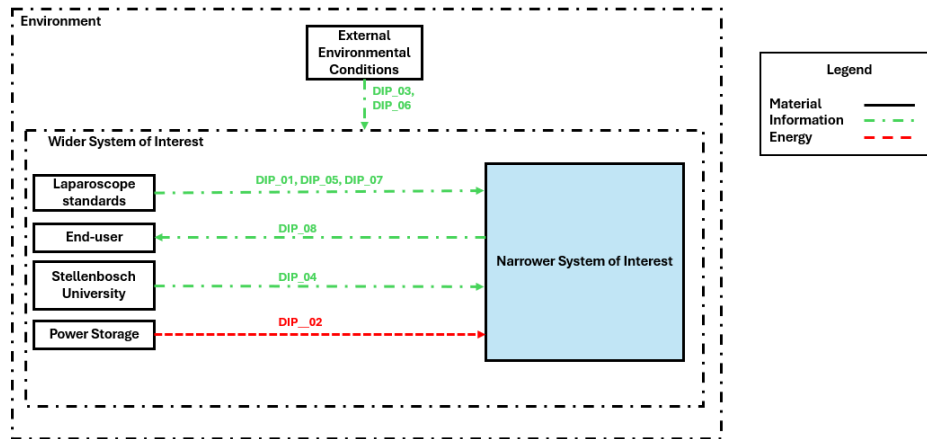


Figure 3.1: System Boundary

Below in *table 3.3* is the functional requirements for the laparoscope. Consequently, the *table 3.4* contains the non-functional requirements.

Table 3.3: Functional Requirements

FR_ID	Functional Requirement Description	Primary/Derived
FR_1	User interface	Primary
FR_2	System processor	Primary
FR_3	Power supply	Primary
FR_4	Operate Camera Module	Primary
FR_4.1	Specify camera settings	Derived
FR_4.2	Specify light intensity	Derived
FR_4.3	Capture digital image	Derived
FR_5.0	Display real-time images	Primary
FR_6.0	Store system data	Primary

Table 3.4: Non-Functional Requirements

NFR_ID	Non-Functional Requirement	Description
NFR_01	Relevant compliance	The design of the system should meet the relevant prescribed standard for this project, to ensure that the laparoscope meets all physical capabilities and functionality of a real laparoscope

Continued on next page...

NFR_ID	Non-Functional Requirement	Description
NFR_02	Documentation	The design, implementation, construction, maintenance and testing procedures for the system should be thoroughly documented
NFR_03	Availability	The components for this project should be available to purchase if a component breaks or should be replaced
NFR_04	Adjustable	The system design should be developed that distinct features can be changed without creating large external costs
NFR_05	Usability	System should be easy to operate for limited-training users

The TPMs can be seen in the *table 3.5* below, based on the values in *table 3.3*. The reasoning behind all the TPM target values can be found below the table.

Table 3.5: Functional Requirements and Technical Performance Measures

FR_ID	Function	TPM_ID	TPM Description	TPM Target Value(s)
FR_1.0	User interface	TPM_01	Minimum user interface input(s)	≥ 1 input
		TPM_02	Minimum user interface signal(s)	≥ 1 signal light
FR_2.0	System processor	TPM_03	Minimum GPIO pins	≥ 1 GPIO Pin
FR_3.0	Power supply	TPM_04	System start up cycle duration	≤ 2 s
		TPM_05	Total duration of power supply required	≥ 6.7 hours
FR_4.0	Operate camera module	TPM_06	Amount of camera modules	1 - 2 camera modules
		TPM_07	Working length of the system	300 mm
		TPM_08	Viewing angle of camera module	30°

Continued on next page...

FR_ID	Function	TPM_ID	TPM Description	TPM Target Value(s)
FR_4.1	Configure camera properties	TPM_09	Focus depth of the camera module	30 mm - 80 mm
FR_4.2	Configure light intensity	TPM_10	Illumination of system	40 000 LUX - 160 000 LUX
		TPM_11	Maximum surface temperature	$\leq 41^{\circ}\text{C}$
FR_4.3	Capture digital image	TPM_12	Minimum field of view	$\geq 70^{\circ}$
		TPM_13	Minimum resolution	460 x 460 pixels
FR_5.0	Display real-time images	TPM_14	Minimum resolution of display	460 x 460 pixels
		TPM_15	Maximum latency	0.5 s
FR_6.0	Store system data	TPM_16	Storage capacity	68.56 GB

- **TPM_01:** The bare minimum for the laparoscope to function properly with user inputs is to at least install a on or off switch. It is the bare minimum required for this project.
- **TPM_02:** For the operator to know the laparoscope is still functioning properly, without looking at the display screen or light module at the end of the laparoscope, there should be at least 1 signal light on the laparoscope to indicate that the system is receiving power.
- **TPM_03:** To operate the signal lights the microcontroller or SBC will require GPIO pin to push current into the signal lights. Thus, the minimum required is the minimum of the laparoscope signal lights.
- **TPM_04:** This is dependent on the component selected and the different interfaces selected between the different components. However the desired start up cycle duration should be less than 2 seconds.
- **TPM_05:** The system should be able to power on for long durations of times. The total amount of time it should be powered on is the duration of a typical laparoscopic surgery.
- **TPM_06:** The system requires at least 1 laparoscopic camera. A second camera module can help the system achieve 3D visualization. Any additional cameras would be redundant for this project.

- **TPM_07:** The system design should be similar to current laparoscopes. Current laparoscope designs has a working distance of 300 mm. The working distance is the distance of the rod that can be inserted into the patient's body.
- **TPM_08:** Different laparoscope products have different viewing angles. Some has a viewing angle of 0°, while there are products that has a viewing angle of 90°. However a common viewing angle is 30°, which is the value that will be pursued in this project.
- **TPM_09:** The distance between the camera sensor of the laparoscope and the object that is being captured is the focus depth.
- **TPM_10:** The total illumination required per medical theatres.
- **TPM_11:** The maximum temperature of the casing of the laparoscope tip, based on medical standards.
- **TPM_12:** Smallest viewing angle applied to a laparoscopic commercial product is 70°, while certain laparoscopes has a viewing angle larger than 70°.
- **TPM_13:** The minimum resolution for the camera module is determined by the first determining the maximum visible seen at the smallest fov, and multiply that value with how many pixels a millimetre area should represented: $\text{Pixel} = \text{Focus depth}_{(\text{max})} \cdot \sin(\text{fov}/2) \cdot 2 \cdot 5 \text{ pixels/mm}$.
- **TPM_14:** The minimum resolution of the display is the same as the minimum resolution of the camera.
- **TPM_15:** This value is subject to change based on the different components used for the system. Therefore, it is important to minimise the latency of each component or interaction between different components. 0.5 s is just an absolute maximum latency that will be accepted.
- **TPM_16:** If the minimum resolution a image is 460 x 460 with RGB colour format, then 1 image requires 634.8 KB. $(P \times P \times 24/8)$ If the frame rate is 30 fps and this continued for 60 minutes, then a total space of 68,5584 Gb is required.

3.2.4 Generate Concept Design for Laparoscope

For this project 3 concepts were generated, based on the principle that it utilized the DF Robot endoscope, which has the dimensions of Ø8 mm and a total length of 25 mm. This module was chosen, since it is the maximum minimum space required to fit all the necessary PCB and camera assembly to create a endoscope, that still fit within the Ø10 mm hole. Each concept is design to have a viewing angle of 30°.

The first concept, relies on bending the shaft that the endoscope camera is attached to. However, the total diameter of the laparoscope within the skin and muscle tissue of the patient should remain under the $\varnothing 10$ mm limit. Since, the thickness of the muscle and tissue of patients are not all the same, the design has to be made for the worst case scenario, which is for a obese patient around 10 cm. *Figure 3.2* below illustrates how the design would look like:



Figure 3.2: Concept 1: Bended shaft design

The second concept, relies that the lens assembly is tilted to 30° relative to the axial axis of the laparoscope. This concept, does require careful hardware consideration and positioning, however is a proven method in the industry. *Figure 3.3* illustrates this design:



Figure 3.3: Concept 2: Tilted lens design

The last concept, relies on a pulley system, which allows the camera to be tilted at a desired angle by the practitioner, while the operation is being performed. This method relies on fine detail design, but can prove to be beneficial for future research purposes. The final concept is illustrate in *figure 3.4* below:



Figure 3.4: Concept 3: Adjustable viewing angle design

All 3 concept mentioned above can be used for the project, since they meet all physical demands of the laparoscope. However, each concepts has their defects. Concept 1 and 3, have both a defect when their camera is tilted at 30°.

When the laparoscope is being rotated, there might be vital information that is missed in the direction of the laparoscopes axial axis, due to the camera lenses not being centred on the axis itself. It can also possibly cause damage to the patients, if the laparoscopes are being turned and accidentally being pushed into the patients internal organs.

Due to the reason stated above, the second concept is the preferred concept that will be further developed.

3.2.5 Finalise Detail Design for Laparoscope

Design and finalise the selected concept design of the laparoscope. This includes producing relevant calculations and analysis of the developed laparoscope and the full drawing pack of the laparoscope.

3.2.6 Compile Design Requirements for Test Station

Determine the requirements for test station of the laparoscope. These requirements include a list of functionality test that should be performable on the test station.

3.2.7 Generate Concept Design for Test Station

Describe, formulate, articulate and design the various tests required for the test station. These tests are to be evaluated and selected based on threshold value they should meet. All the selected tests will then be implemented in a final test station concept, which upon completion would be evaluated.

3.2.8 Finalise Detail Design for Test Station

Articulate and design the final test station for the laparoscope. This includes producing a safety report of the test station, instruction documents of how the various test should be performed and a full drawing pack of the test station.

3.2.9 Construct and Implement Laparoscope

Construction of the developed laparoscope should be conducted. This includes all software implementation and building of the developed product.

3.2.10 Construct Test Station

Construct the test station for the laparoscope. This entails all building requirements of the test station.

3.2.11 Design and Perform Functionality Test

Design and perform the functionality tests of the built laparoscope. This includes following all test instructions and post processing of the results.

3.2.12 Evaluate Laparoscope

Evaluate the performance of the developed laparoscope.

3.2.13 Documentation

Document the entirety of the project. All relevant literature review, design specifications, concept designs, test procedures, results and evaluation should be thoroughly documented. At predetermined dates, it is required to produce relevant documents, including the progress report, preliminary report and final report.

3.3 Resource Requirements

There are various equipment, components and lab access required to complete this project. As mentioned in the literature review's section 2.4, there are 3 main hardware components required for this project. However, there are various hardware components required for this project:

- Single board computers.
- Camera module.
- Illumination module.
- Display screen or monitor.
- Power supply.
- Electrical components.(Resistors, Light-emitting diode(LED), wires, buttons)

There will also be required to manufacture a selection of parts for the laparoscope and test station. These parts will mainly consist of 3D printed parts and 1 thin walled rod with a diameter of less than 10 mm.

There will be access required to the mechanical and mechatronic laboratories, to build, program and test the final product.

3.4 Expected Results

The expected results of this project, is to have a low-cost, functional laparoscope at the end of the project. To classify as a low-cost, functional laparoscope, the laparoscope should meet these requirements:

- The laparoscopes rod and camera assembly should fit within a 10 mm diameter hole.
- The captured real-time digital images should allow the operator to accurately coordinate the equipment.
- During testing, the laparoscope should meet a minimum required pass rate of 80%.

3.5 Project Schedule

In *figure 3.5* below, a gantt chart is shown to illustrate the project schedule.

Gantt Chart

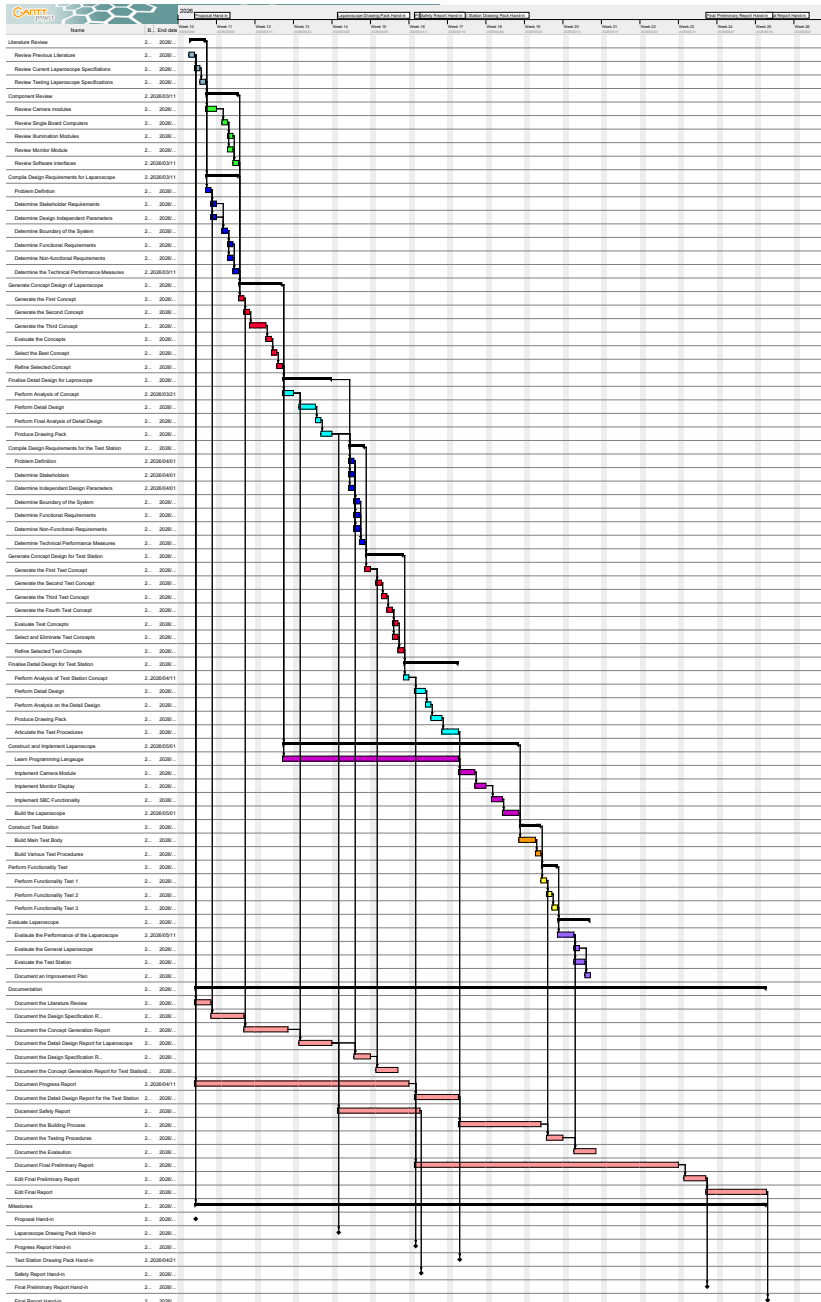


Figure 3.5: Skripsie Gantt Chart

3.6 Risk Assessment

This section of the proposal will cover all relevant risks of this project. Risk is an important feature of any project that needs to be maintained and mitigated to prevent the project from failing.

The first class of risk that will be discussed in this project is that of safety. There are various risk associated with safety problems, for the project and the researcher. These risk mainly occur during building and testing phases. Hence, when the researcher works in the laboratories. These risks include health risks for the researchers and potential damages to the project, leading to time delay to receive replacement parts. To mitigate these risks it will be required to write a safety report, that needs to be reviewed and signed, by the researcher's supervisor and laboratories manager. These safety reports will always be kept on the researcher during laboratories sessions and will required to be completely followed.

Product delay and delivery times is a risk that needs to be handled carefully. Due to the short duration of this skripsi (6 months), product delivery times is a risk to the completion time of this product. If products are required to be imported from international suppliers, then there is a risk that the project could become behind schedule. The only method to mitigate this is by using local suppliers and to order products in a timely manner. If the international products are required, then the products should be order at the earliest convenience. Another mitigation method is to order products local suppliers currently have in stock. It is essential that design critical components are handled with care, to prevent damage to the components, which would require replacement components.

The last class associated with this project is lack of access to the laboratories. This can occur due to several reasons, specifically if there isn't laboratories availability due to limited space or construction of the mechanical and mechatronic department. To minimize this risk, access to the laboratories will be asked at earliest convenience. If access to the mechanical and mechatronic laboratories are unavailable, then access to electrical and electronic departments laboratories will be queried.

Chapter 4

Conclusions

4.1 Conclusion

Laparoscopy is a rapidly expanding industry in the medical field. Its an innovative approach to minimize the harm surgeons inflict on the patients. However, it is a more complex procedure to perform, thus requiring more training. Consequently, there is a desire for medical robots to assist surgeons during laparoscopic surgery.

Therefore, there is a need for a low-cost laparoscope for training purposes. Consequently, it can be used for prototype testing of new medical robots. This low-cost laparoscope is to be tested to determine its functionality. Therefore, a testing station is required to be developed to determine the laparoscope's capabilities.

For this reason, this project aims to develop a low-cost laparoscope in addition to a testing station.

List of references

- arducam.com (2020). Arducam 12mp*2 synchronized stereo camera bundle kit for raspberry pi and pi zero, two 12.3mp 477p camera modules with cs lens and camarray stereo camera hat. Tech. Rep., ArduCam.
Available at: <https://www.arducam.com/arducam-12mp2-synchronized-stereo-camera-bundle-kit-for-raspberry-pi-two-12-3mp-imx477-camera-modules-with-cs-lens-and-arducam-camarray-stereo-camera-hat.html>
- Buia, A., Stockhausen, F. and Hanisch, E. (2015). Laparoscopic surgery: A qualified systematic review. *World J Methodol*, vol. 5, no. 4, p. 238–54.
- cambridgeincolour.com (2020a). Camera exposure. Tech. Rep., Cambridge in Colour.
Available at: <https://www.cambridgeincolour.com/tutorials/camera-exposure.htm>
- cambridgeincolour.com (2020b). Digital camera sensors. Tech. Rep., Cambridge in Colour.
Available at: <https://www.cambridgeincolour.com/tutorials/camera-sensors.htm>
- docs.arduino.cc (2026). Language reference. Tech. Rep., Arduino AG.
Available at: <https://docs.arduino.cc/language-reference/#structure>
- ebme.co.uk (2026). Endoscope design and assembly. Tech. Rep., EBME and Clinical Engineering Articles.
Available at: <https://www.ebme.co.uk/articles/clinical-engineering/endoscope-design-and-assembly>
- Ibrahim, A., Ulrich, M., Liselotte, M., Julian, P., Nicolai, M., Matthias, B., Georgios, G., Antonio Simone, L. and Damaris, F. (2021). The development of laparoscopy—a historical overview. *Frontiers in surgery*, vol. 8.
- intuitive.com (2026). Da vinci sp. Tech. Rep., Intuitive Surgical.
Available at: <https://www.intuitive.com/en-us/products-and-services/da-vinci/sp>
- karlstorz.com (2026a). Home page. Tech. Rep., Karl Storz - Endoskope.
Available at: <https://www.karlstorz.com/de/en/index.htm>
- karlstorz.com (2026b). Tipcam1 rubina. Tech. Rep., Karl Storz - Endoskope.
Available at: <https://www.karlstorz.com/za/en/product-detail-page.htm?>

productID=1000108799&cat=1000122034&etcc_cu=onsite&etcc_med_onsite=Search-OnlineCatalog-Overlay&etcc_cmp_onsite=Search-Overlay&etcc_st_onsite=Tipcam%20Rubina

leopardimaging.com (2026). Hummingbird mobile camera modules. Tech. Rep., Leopard Imaging.

Available at: <https://leopardimaging.com/hummingbird/>

Morrow, D. (2026). Ultimate depth of field photography guide [2026]. Tech. Rep., Dave Morrow Photography.

Available at: <https://www.davemorrowphotography.com/depth-of-field-photography>

olympus.com.au (2026). Endoeye 3d 10 mm. Tech. Rep., Olympus Corporation.

Available at: <https://www.olympus.com.au/medical/en/Products-and-Solutions/Products/Product/ENDOYE-3D-10-mm.html>

Public (2022). Arducam/arduino. Tech. Rep., Github.com.

Available at: <https://github.com/ArduCAM/Arduino>

raspberrypi.com (2026a). Documentation. Tech. Rep., Raspberry Pi Ltd.

Available at: <https://www.raspberrypi.com/documentation/>

raspberrypi.com (2026b). Products. Tech. Rep., Raspberry Pi Ltd.

Available at: <https://www.raspberrypi.com/products/>

robotics.org.za (2026). Home page. Tech. Rep., Micro Robotics (Pty) Ltd.

Available at: <https://www.robotics.org.za/>

store.arduino.cc (2026). Products. Tech. Rep., Arduino AG.

Available at: <https://store.arduino.cc/products>

Subido, Edwin D. C., J., Pacis, D.M.M. and Bugtai, N.T. (2018). Recent technological advancements in laparoscopic surgical instruments. *AIP Conference Proceedings*, vol. 1933, no. 1.

visionmeta.com (2026). 1mp 3.3/3.5/3.9mm ov9734 hd endoscope medical camera module. Tech. Rep., Shenzhen VisionMeta Technology Co., Ltd.

Available at: <https://www.visionmeta.com/product-category/feature/endoscope/endoscope-diameter/3-5mm-endoscope/>

Xiao, D., Jakimowicz, J.J., Albayrak, A., Buzink, S.N., Botden, S.M.B.I. and Goossens, R.H.M. (2014). Face, content, and construct validity of a novel portable ergonomic simulator for basic laparoscopic skills. *Journal of surgical education*, vol. 71, no. 1, p. 65–72.

Estimated Cost per Activity

The estimated cost per activity, shows all the planned activities of the project and how much time would be used to perform each activity. Each activity duration is determined by the gantt chart. It is assumed that there is 8 hours available each working day. And the workload is divided based on importance for each day. Usually 5 hours for planned activities and then 1 hour each day for documentation. This is not a concrete value and is subject to change.

Table .1: Estimated Cost per Activity

Activity	Engineering Time		Running Cost	Facility Use	Capital Cost	MMW			Total
						Labour	Material		
	hr	R	R	R	R	hr	R	R	R
Literature Review	15	6750	150						6900
Component Review	20	9000	150						9150
Compile Design Requirements for Laparoscope	20	9000	150						9150
Generate Concept Design for Laparoscope	35	15750	150						15900
Finalise Detail Design for Laparoscope	40	18000	150						18150
Compile Design Requirements for Test Station	20	9000	150						9150
Generate Concept Design for Test Station	30	13500	150						13650
Finalise Detail Design for Test Station	40	18000	150						18150
Constuct and Implement Laparoscope	95	42750	600	6000	4000	15	4500	500	58350
Construct Test Station	20	9000	600	2000		40	12000	300	23900
Perform Functionality Test	40	18000	600	2000					20600
Evaluate Laparoscope	25	11250	150						11400
Documentation	120	54000	150						54150
Total	520	234000	3300	10000	4000	55	16500	800	268600

GA Self Assessment

Graduate Attribute	How it will be satisfied
GA 1 - Problem solving: Identify, formulate, research literature and analyse complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences and engineering sciences with holistic considerations for sustainable development.	This GA will be satisfied during the literature review, design specification reports, concept generation reports and the detail design reports, including drawing packs.
GA 2 - Application of scientific and engineering knowledge: Apply knowledge of mathematics, natural science, computing and engineering fundamentals, and an engineering specialisation to develop solutions to complex engineering problems.	GA 2 will be satisfied during sections where component specification and selection will be conducted. It will also be satisfied during detail design sections where analysis of the design will be performed.
GA 3 - Engineering Design: Design creative solutions for complex engineering problems and design systems, components or processes to meet identified needs.	This GA will be satisfied by following the workplan and planned schedule of the project. It will also be satisfied during component selection, detail design process and the evaluation of the developed product.

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Graduate Attribute (continued)	How it will be satisfied (continued)
GA 5 - Use of Engineering Tools: Demonstrate competence to create, select and apply and recognise limitations of appropriate techniques, resources and modern engineering and IT tools, including prediction and modelling, to complex engineering problems.	GA 5 will be satisfied during the detail design of the project. It will also be satisfied during the implementation of the laparoscope.
GA 6 - Professional and technical communication: Demonstrate competence to communicate effectively and inclusively on complex engineering activities, both orally and in writing, with the engineering community and society at large, taking into account cultural, language, and learning differences.	GA 6 will be satisfied during the documentation of the project, including the progress and final report. It will also be satisfied during the weekly meetings, progress meeting and final presentation.
GA 8 - Individual and collaborative teamwork: Demonstrate competence to function effectively as an individual and as a member or leader in diverse and inclusive teams, and in multidisciplinary, face-to-face, remote and distributed settings.	This GA will be satisfied during the construction of the laparoscope and testing station. It will also be satisfied during the testing stages of the laparoscope.
GA 9 - Independent Learning Ability: Demonstrate competence to engage in independent learning through well-developed learning skills.	This will be satisfied during the entirety of the project, but predominantly during the literature review.

Table .2: GA self-assessment